Chapter 11:
Electricity

Topic Outline

- Electric Current, Voltage, Resistance
- Resistivity
- Ohm’s Law
- Power

Learning Outcomes

- Define current, resistance and resistivity
- Identify resistors through color codes
- Define Ohm's law
- Mathematically relate the three basic quantities, current, voltage and resistance through the concept of function
- Recognize the common electrical symbols used to draw circuit diagrams

Electricity

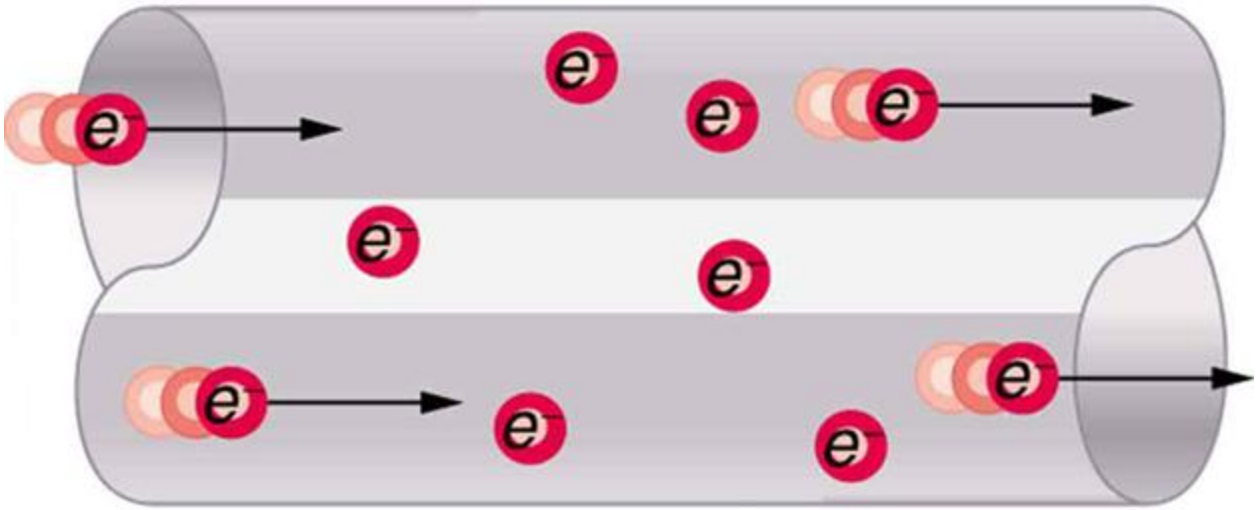
- Phenomenon associated with stationary or moving electrical charges

Electricity

- the flow of electrical charge around a circuit generates electricity

Electricity

- The movement of charge around a closed circuit in the form of electrons → **electric current**



Electricity

charge without any flow or movement is called static electricity

Current

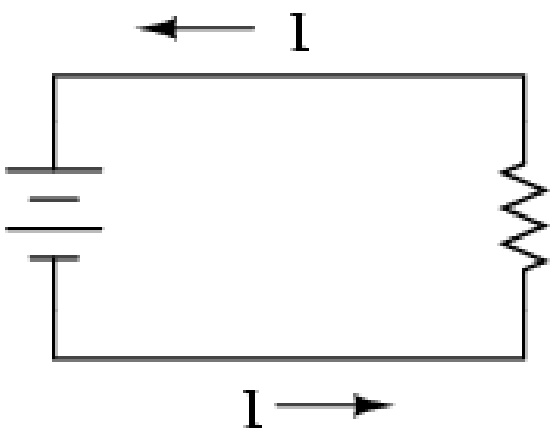
- Current (I) – the rate of flow of charge through the conductor

$$I = q / t$$

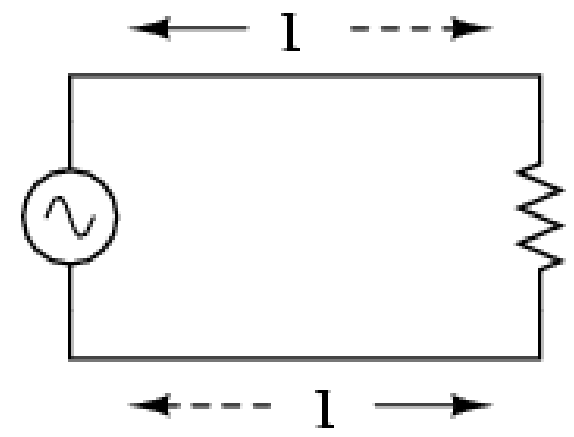
Unit: Ampere = Coulomb/second

$$A = C/s$$

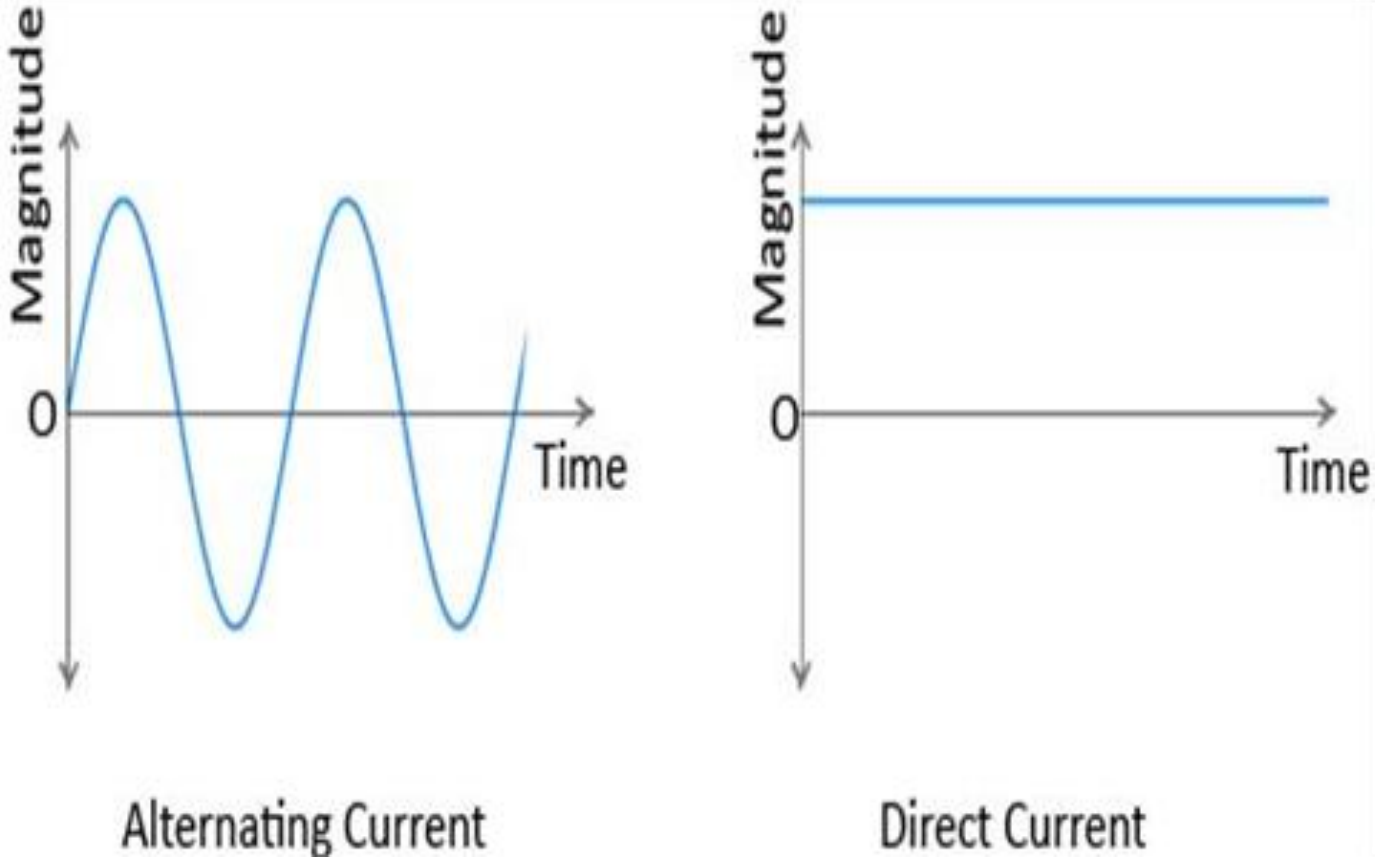
DIRECT CURRENT
(DC)



ALTERNATING CURRENT
(AC)



Current


Magnitude
0
Time
Alternating Current
Magnitude
0
Time
Direct Current

Voltage

- Voltage (V) - is the potential energy of an electrical supply stored in the form of an electrical charge.

Voltage

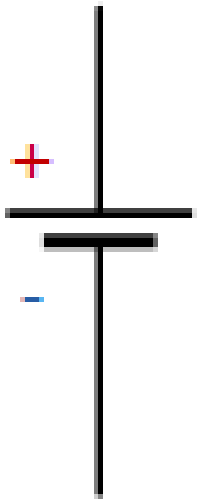
- the electrical energy needed for a charge to flow through the conductor

Voltage = Energy / charge

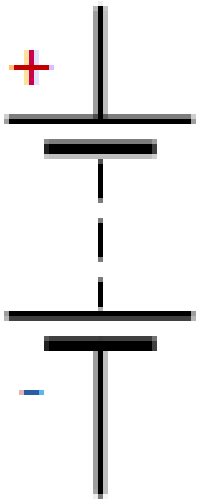
unit: Volt = Joules / Coulomb

V = J / C

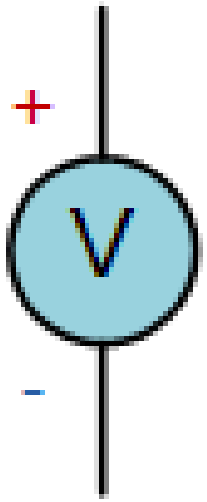
Voltage



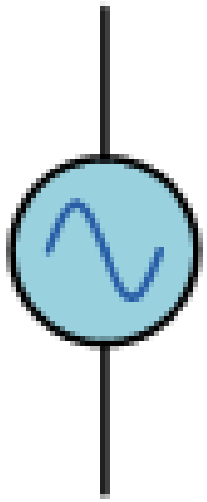
Single Cell



Multiple Cells (Battery)



DC Voltage Source



AC Voltage Source

Batteries


Li-ion Battery
3.7V 500 mAh


AA 700mAh 1.2V




MOTOLITE

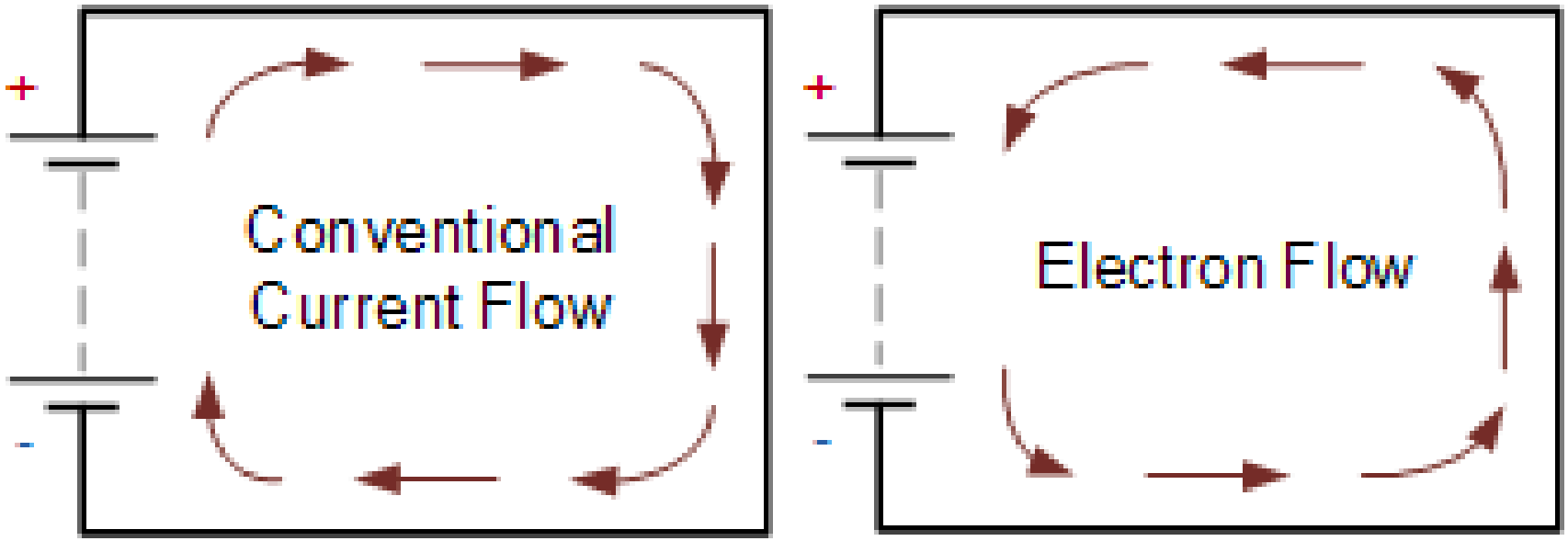

+ | AA
12-2024
Energizer


KAYO Ni-MH
AA 1600mAh 1.2V

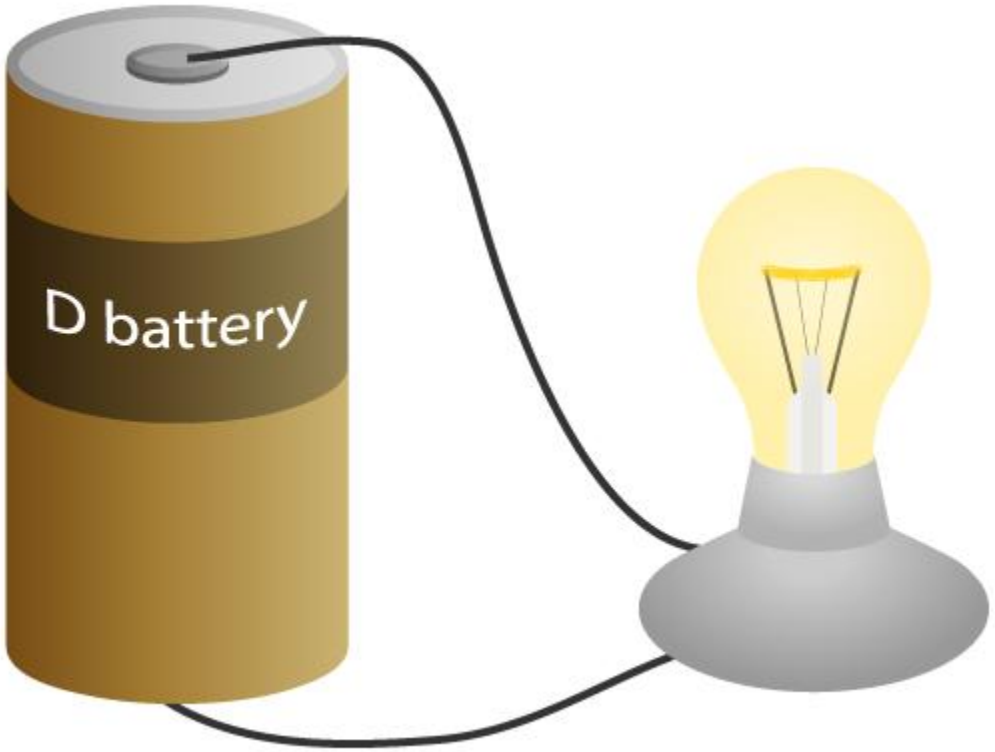
KAYO Ni-MH
AA 1600mAh 1.2V


Lithium Cell
CR1220
3V

Electron Flow



Voltage



Voltage is also known as potential difference

Positive terminal of a battery has the **higher potential energy**

Resistance

- Resistance (R) – is a measure of the opposition to current flow in an electrical circuit

Resistors

- Resistors can be made from a variety of materials, carbon, metal oxide, metal film, resistance wire and the like.

Resistors

- Resistors can come in different formats - different types of resistor have slightly different characteristics and this means that they may be used in different circuit applications.

Types of Resistors



Carbon composition Resistors



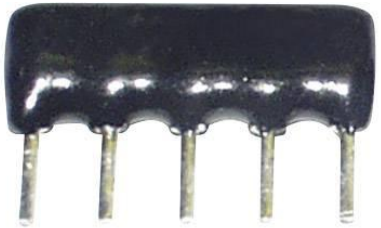
Carbon film resistors



Metal film Resistors



Wire wound resistors



Cermet film resistors

Types of Resistors



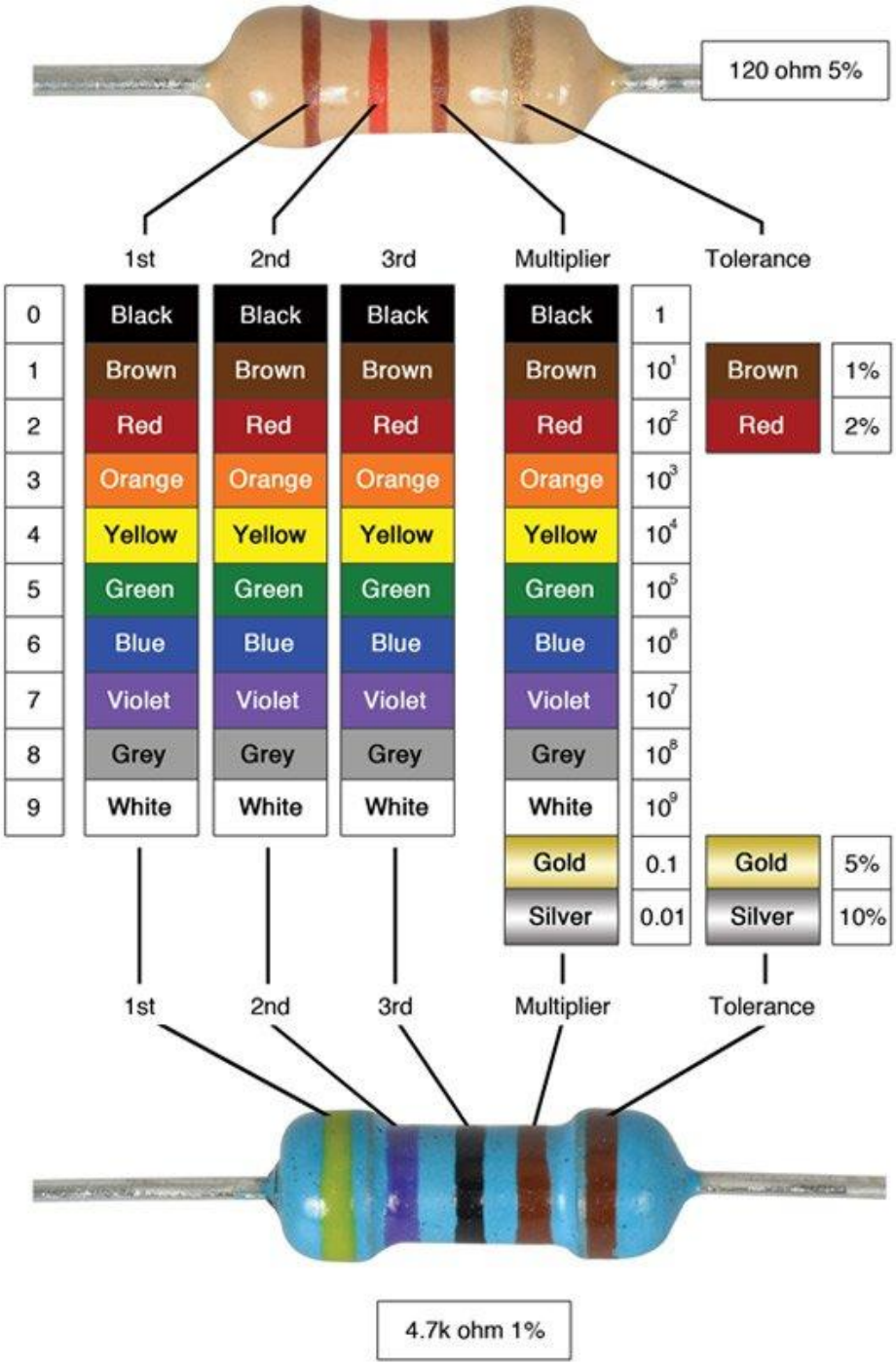
Rheostat



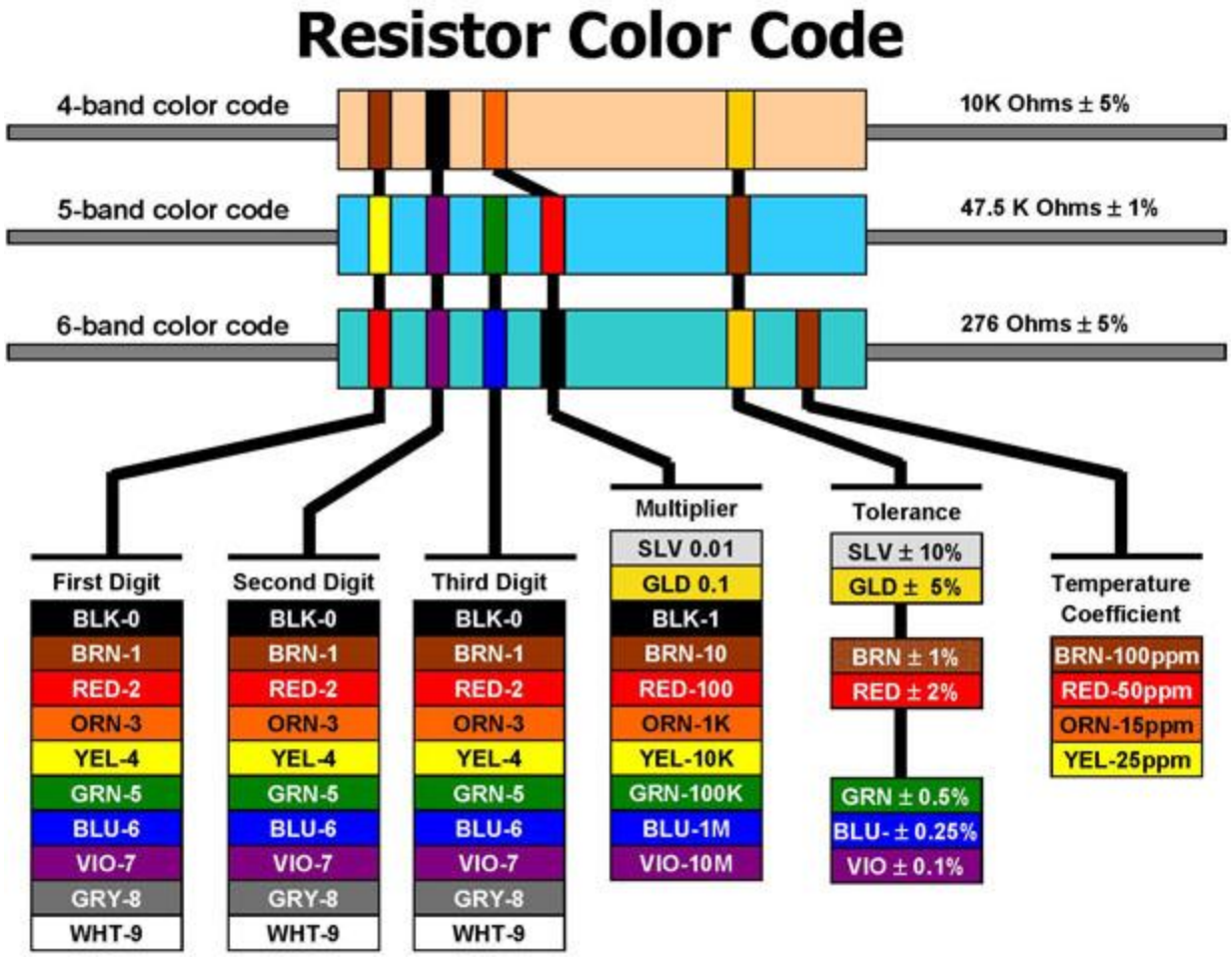
Potentiometer

Color Code




120 ohm 5%
1st
2nd
3rd
Multiplier
Tolerance
0 Black Black Black Black 1
1 Brown Brown Brown Brown 10^1 Brown 1%
2 Red Red Red Red 10^2 Red 2%
3 Orange Orange Orange Orange 10^3
4 Yellow Yellow Yellow Yellow 10^4
5 Green Green Green Green 10^5
6 Blue Blue Blue Blue 10^6
7 Violet Violet Violet Violet 10^7
8 Grey Grey Grey Grey 10^8
9 White White White White 10^9
Gold 0.1 Gold 5%
Silver 0.01 Silver 10%
1st
2nd
3rd
Multiplier
Tolerance
4.7k ohm 1%

Resistors


Resistor Color Code
4-band color code
10K Ohms ± 5%
5-band color code
47.5 K Ohms ± 1%
6-band color code
276 Ohms ± 5%
First Digit
BLK-0
BRN-1
RED-2
ORN-3
YEL-4
GRN-5
BLU-6
VIO-7
GRY-8
WHT-9
Second Digit
BLK-0
BRN-1
RED-2
ORN-3
YEL-4
GRN-5
BLU-6
VIO-7
GRY-8
WHT-9
Third Digit
BLK-0
BRN-1
RED-2
ORN-3
YEL-4
GRN-5
BLU-6
VIO-7
GRY-8
WHT-9
Multiplier
SLV 0.01
GLD 0.1
BLK-1
BRN-10
RED-100
ORN-1K
YEL-10K
GRN-100K
BLU-1M
VIO-10M
Tolerance
SLV ± 10%
GLD ± 5%
BRN ± 1%
RED ± 2%
GRN ± 0.5%
BLU- ± 0.25%
VIO ± 0.1%
Temperature Coefficient
BRN-100ppm
RED-50ppm
ORN-15ppm
YEL-25ppm

Ohm’s Law

Georg Simon Ohm



- German physicist
- Conceptualized the relationship among current, voltage and resistance

Ohm's Law

The current flowing through an electrical is directly proportional to the potential difference (voltage) and inversely proportional to the resistance.

$$I \propto V$$

$$I \propto 1/R$$

$$I \propto V/R$$

$$I = V/R$$

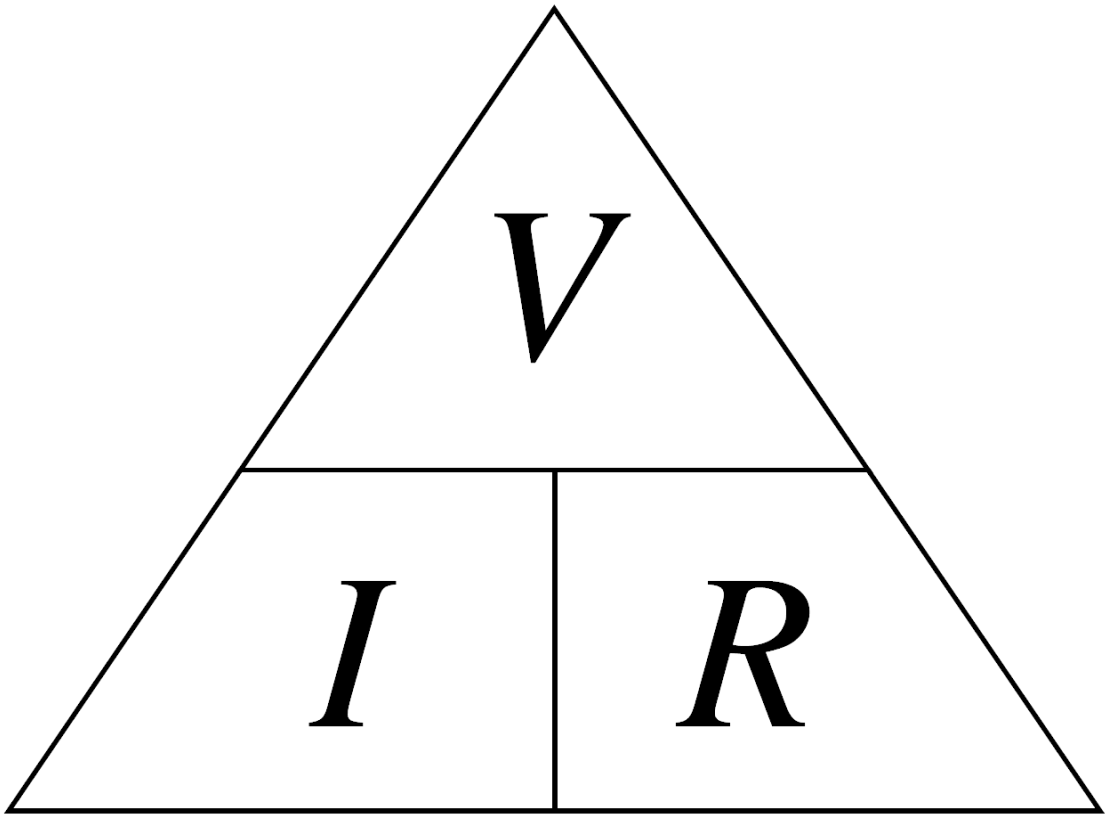
Ohm's Law



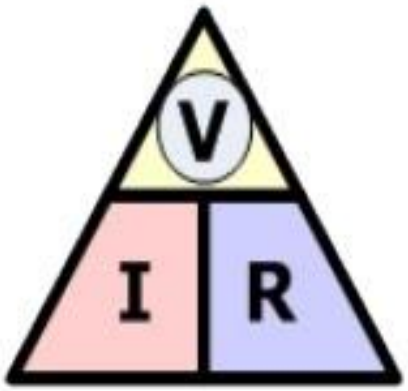
Voltage:

Water pressure

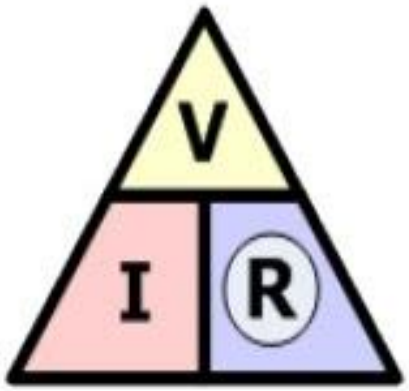
Ohm’s Law



Ohm's Law Triangle



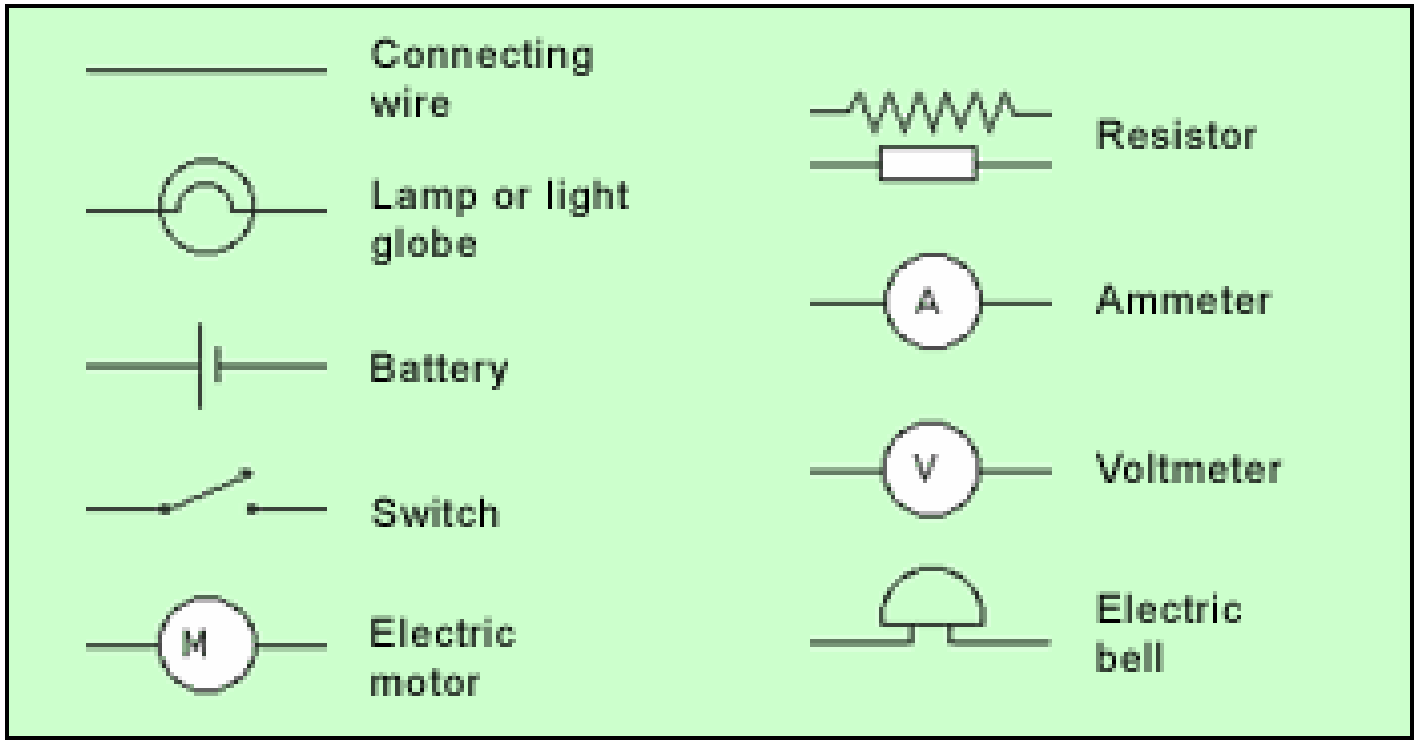




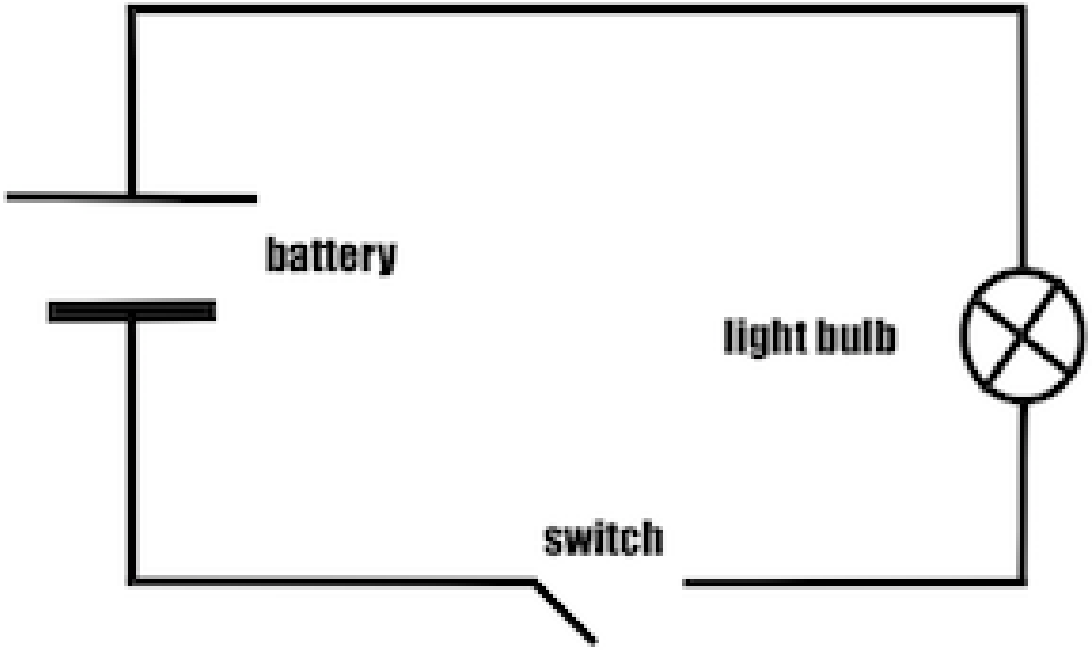
$V = I \times R$ $\quad I = \frac{V}{R}$ $\quad R = \frac{V}{I}$

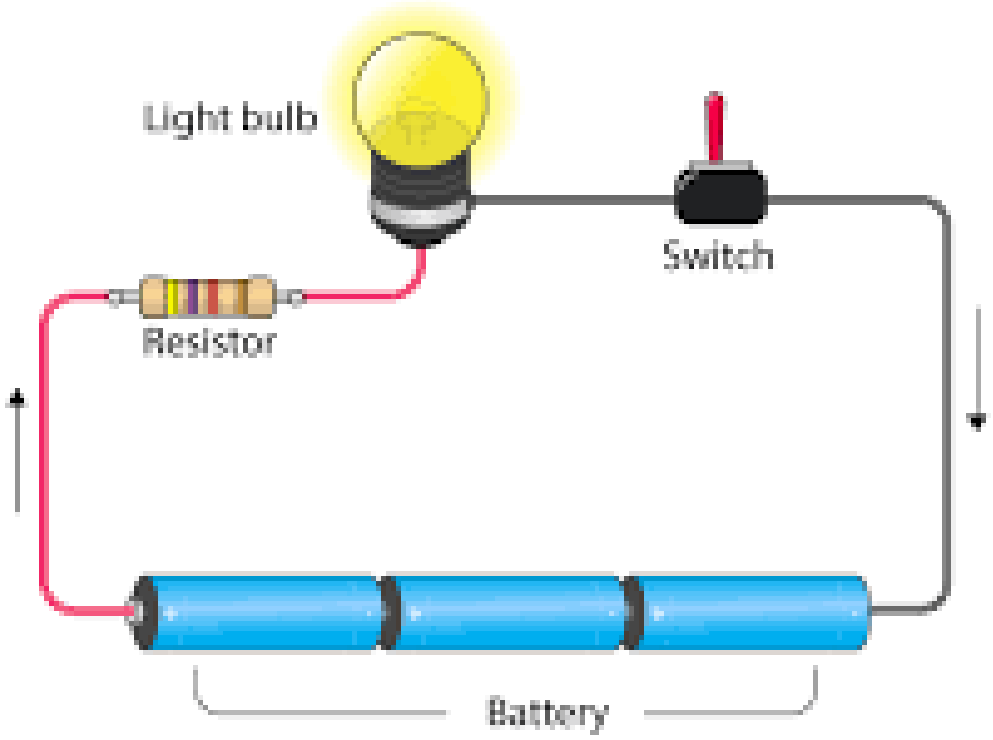
Circuit Diagram

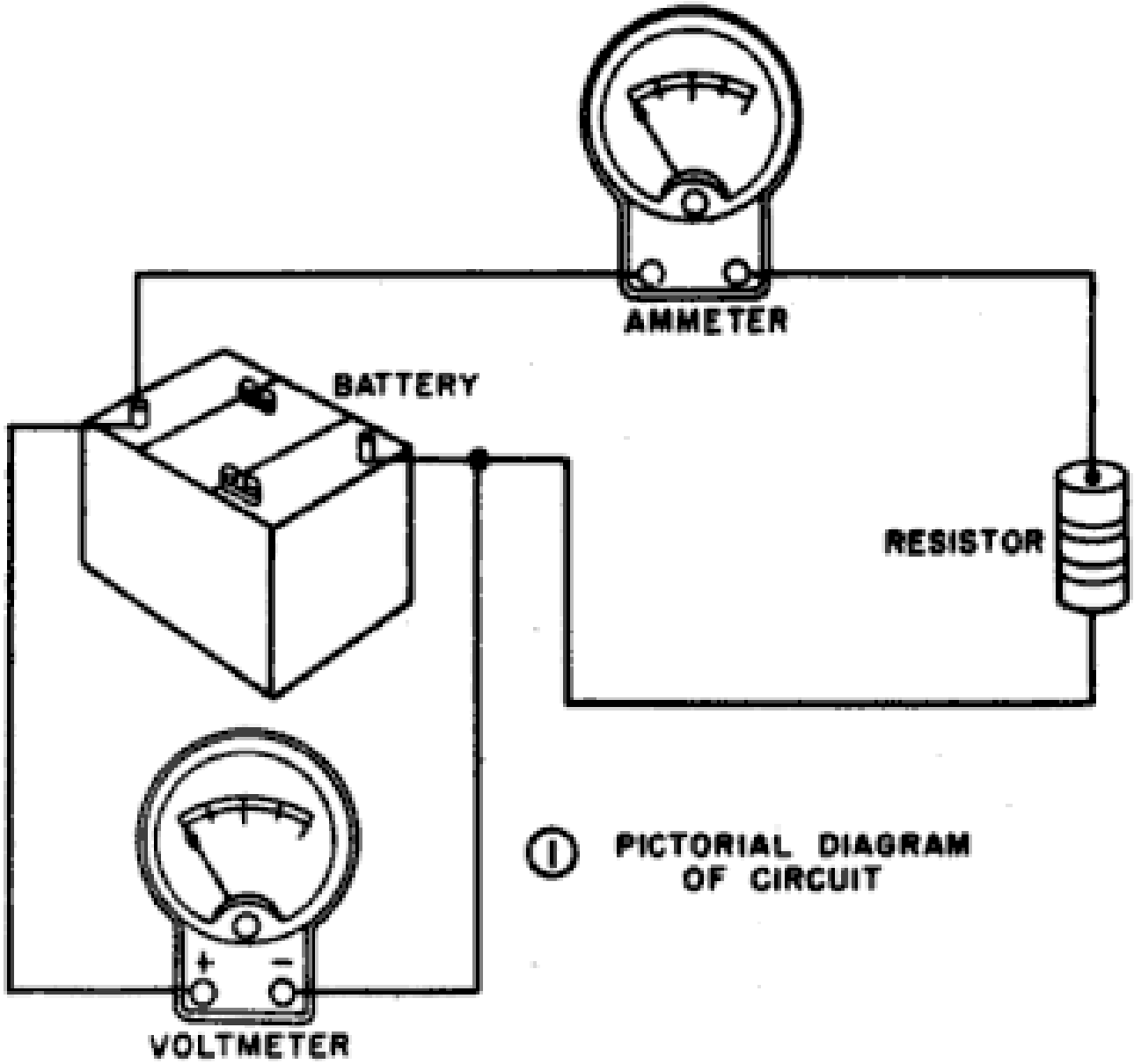
Circuit Diagram Components



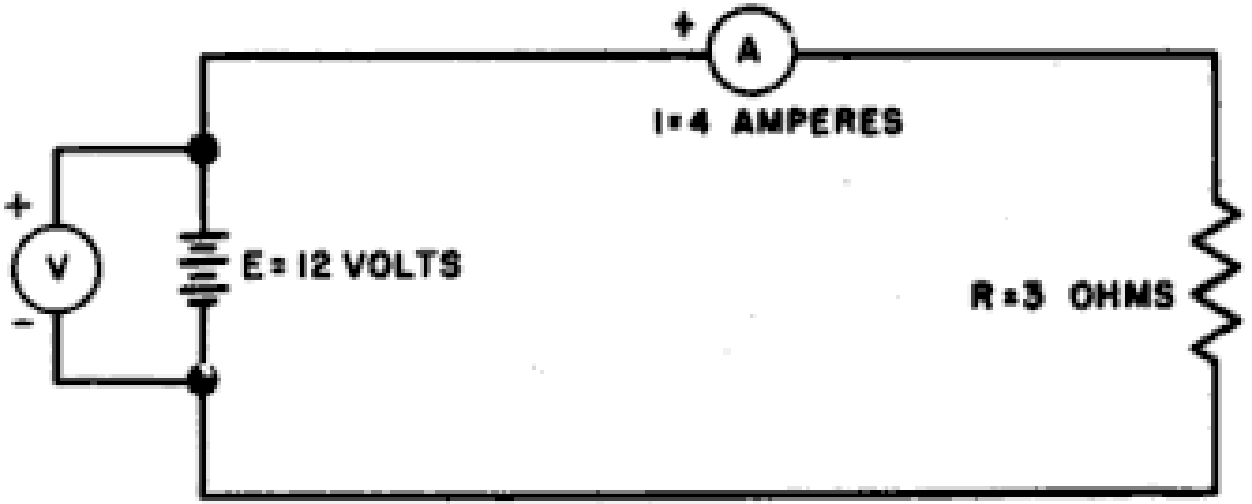
Simple Electric Circuit




Light bulb
Switch
Resistor
Battery



① PICTORIAL DIAGRAM OF CIRCUIT



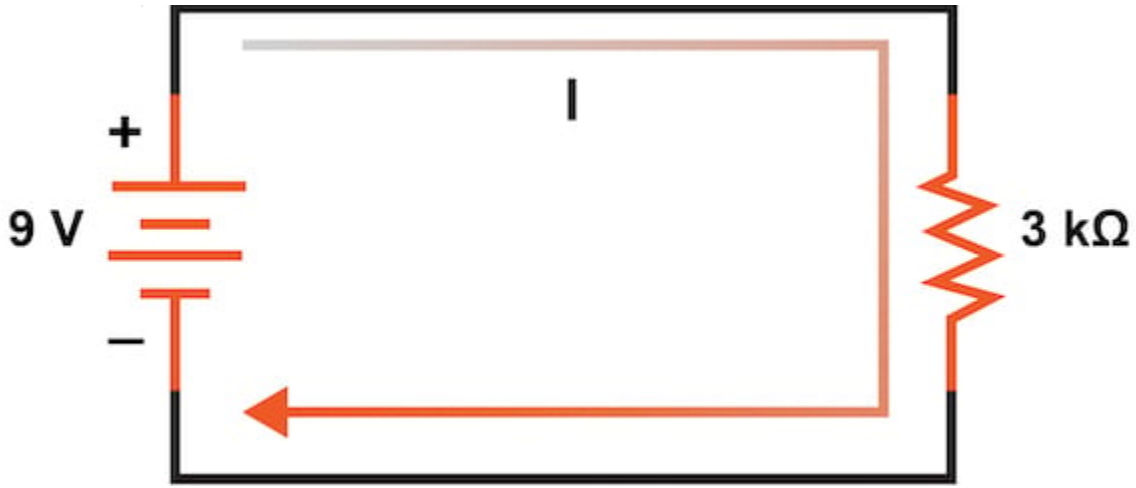
② SCHEMATIC OF CIRCUIT

Sample Problem #1

Calculate the current given the circuit diagram.



Calculate the current given the circuit diagram.



I = V/R

= 9 V / 3,000 Ω

I = 3×10^{-3} A or 3mA

Resistivity

Resistivity

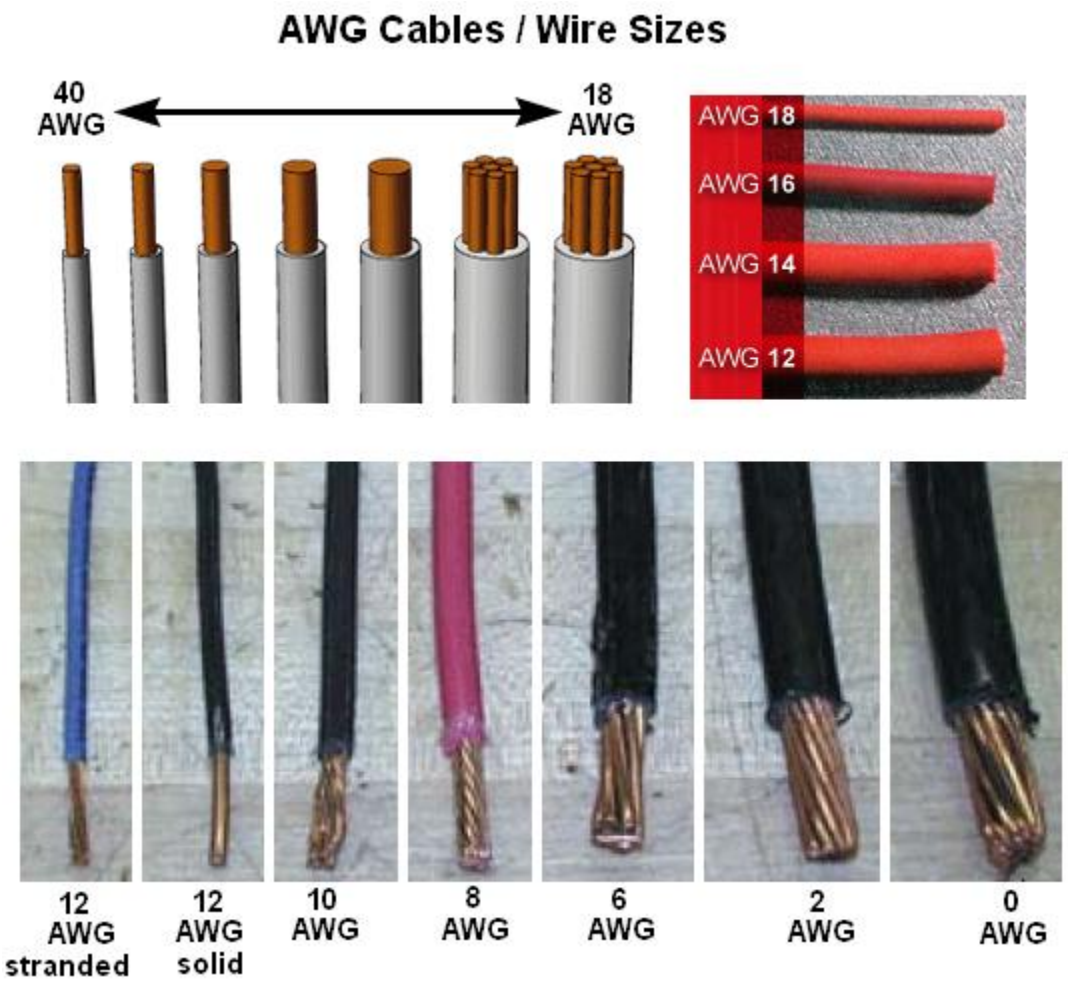
- is a property that describes the extent to which a material opposes the flow of electric current through it.

Resistance





Resistance


AWG Cables / Wire Sizes
40 AWG
18 AWG
AWG 18
AWG 16
AWG 14
AWG 12
12 AWG stranded
12 AWG solid
10 AWG
8 AWG
6 AWG
2 AWG
0 AWG

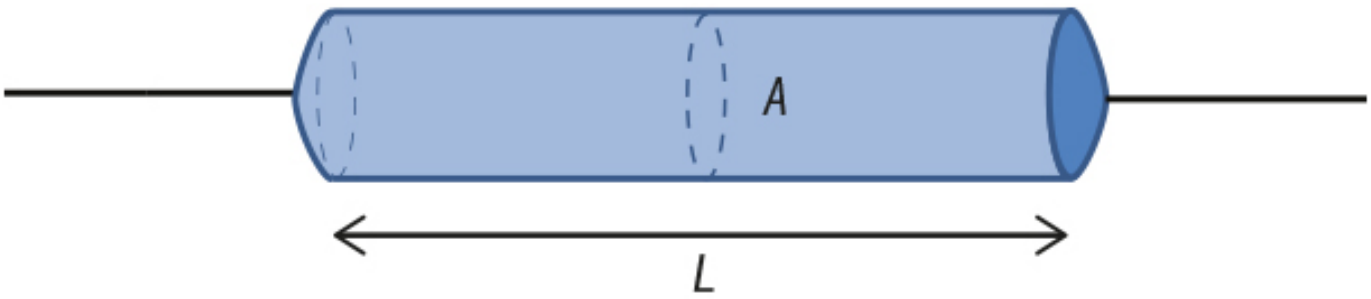
American Wire Gauge (AWG)	Diameter (inches)	Diameter (millimeters)
1	0.289	7.35
2	0.258	6.54
3	0.229	5.83
4	0.204	5.19
5	0.182	4.62
6	0.162	4.11
7	0.144	3.67
8	0.129	3.26
9	0.114	2.91
10	0.102	2.59
11	0.0907	2.30
12	0.0808	2.05
13	0.0720	1.83
14	0.0641	1.63
15	0.0571	1.45
16	0.0508	1.29
17	0.0453	1.15
18	0.0403	1.02
19	0.0359	0.91
20	0.0320	0.81
21	0.0285	0.72
22	0.0254	0.65
23	0.0226	0.57
24	0.0201	0.51
25	0.0179	0.45
26	0.0159	0.40
27	0.0142	0.36
28	0.0126	0.33
29	0.0113	0.28
30	0.0100	0.25
31	0.0089	0.22
32	0.0080	0.20

Resistance



Wires come in different sizes. The maximum current each size can conduct safely is shown.

Resistance



$R \propto L$

$R \propto 1/A$

$R \propto L/A$

$R = \rho L/A$

Resistance

$$R = \rho L/A$$

Resistivity

$$\rho = RA/L$$

Temperature Dependence of Resistance

- Electrons flowing through a conductor are impeded by atoms and molecules. The more these atoms and molecules bounce around, the harder it is for the electrons to get by.

Temperature Dependence of Resistance

- The resistance of most conductors increases as temperature rises because higher thermal energy causes atoms to vibrate more, hindering electron flow.

Temperature Dependence of Resistance

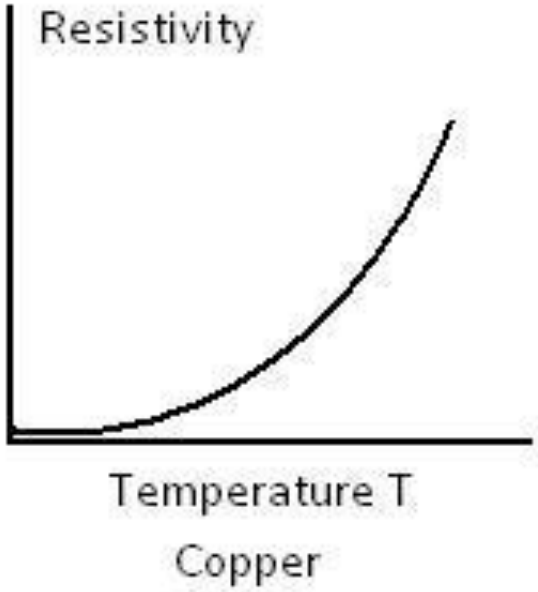
- Resistance generally increases with temperature.

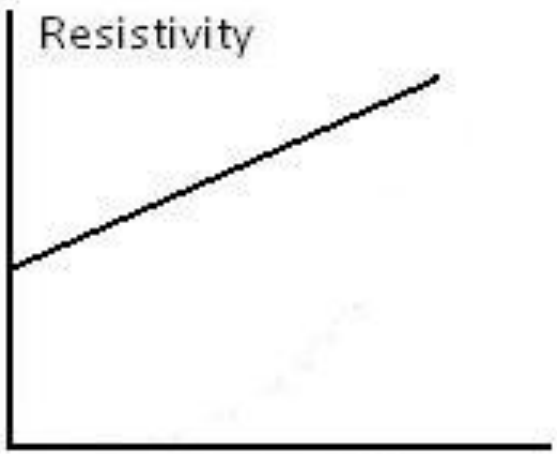
$$R = R_o (1 + \alpha \Delta T)$$

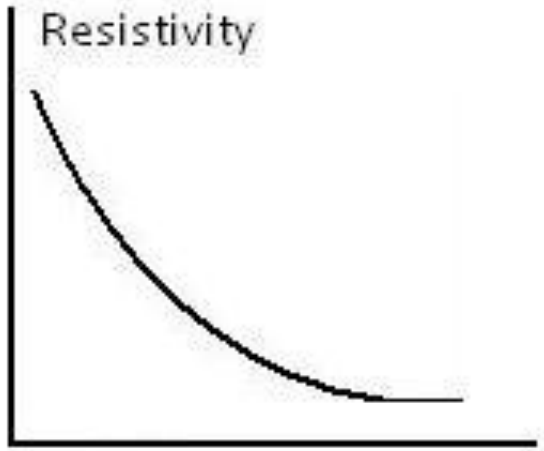
α – temperature coefficient of resistance

Temperature Dependence of Resistance

- A positive temperature coefficient results in resistance increasing with temperature
- A negative temperature coefficient results in a decreasing resistance with an increasing temperature.







- Metal (Cu) – graph of resistivity and temperature is not linear
- Alloy – like Nichrome
- Semiconductor- negative temperature coefficient

Sample Problem #2

If the resistance is 1.50Ω, ρ = 2.65x10-8 Ωm and the length is 100m, what is the radius of the wire?

If the resistance is 1.50 Ω, resistivity is equal to 2.65×10^{-8} Ωm and the length is 100m, what is the radius of the wire?

Given:
R = 1.50 Ω
$\rho = 2.65 \times 10^{-8}$ Ω m
L = 100 m

Required:
radius (r)

Equation:
$R = \rho L / A$

Solution:
$R = \rho L / A$

$A = \rho L / R$

$= (2.65 \times 10^{-8}\ \Omega\ m)(100\ m) / 1.50\ \Omega$

$A = 1.77 \times 10^{-6}\ m^2$ or $1.77\ mm^2$

$A = \pi r^2$

$r = \sqrt{A / \pi}$

$= \sqrt{1.77\ mm^2 / \pi}$

r = 0.75 mm

Sample Problem #3

A platinum resistance thermometer has a resistance of 100.0 Ω at 20°C. If the temperature coefficient of resistance for platinum is 0.00392 K^{-1}, what is the resistance at 50°C?

A platinum resistance thermometer has a resistance of 100 Ω at 20°C. If the temperature coefficient of resistance for platinum is 0.00392 K^{-1}, what is the resistance at 50°C?

Given:

R_o = 100 Ω

α = 0.00392 K^{-1}

Ti = 20°C

Tf = 50°C

Required:

R

Equation:

$$R = R_o (1 + \alpha \Delta T)$$

Solution:

$$R = R_o (1 + \alpha \Delta T)$$

$$= 100\ \Omega\ [1 + 0.00392/K\ (50°C - 20°C)]$$

R = 111.76 Ω

Power

Electric Power

- the rate of electrical energy transfer by an electric circuit per unit of time

Power = Energy / time

P = E / t

Unit: J / s = Watt

Electric Power

$P = VI$

unit:

$= (J/C)\ (C/s)$

$= J/s$

$= Watt$

From Ohm's law

$P = VI$

$P = (IR)\ I$

$P = I^2R$

$P = VI$

$P = V\ (V/R)$

$P = V^2/\ R$

Sample Problem #4

A 1.5 hp air-conditioning unit is used daily for a period of 8 hours. What is the amount of the energy consumption for a 30 day-month if the Meralco's electricity rate is around ₱13.82 per kWh.

A 1.5 hp air-conditioning unit is used daily for a period of 8 hours. What is the amount of the energy consumption for a 30 day-month if the Meralco's electricity rate is around ₱13.82 per kWh.

Given:
P = 1.5 hp
t= 8 h /day
electricity rate = ₱13.82 / kWh

Required:
Amount energy consumption

Equation
P = E / t

Solution:
E = P t

Solution:
E = P t
= 1.5 hp (746 W / hp) [8 h /day (30 day)]
E = 268,560 Wh
E = 268.56 kWh

Amount = Energy x rate
= 268.56 kWh x ₱13.82 / kWh
Amount = ₱ 3,711.50

End of Part 1